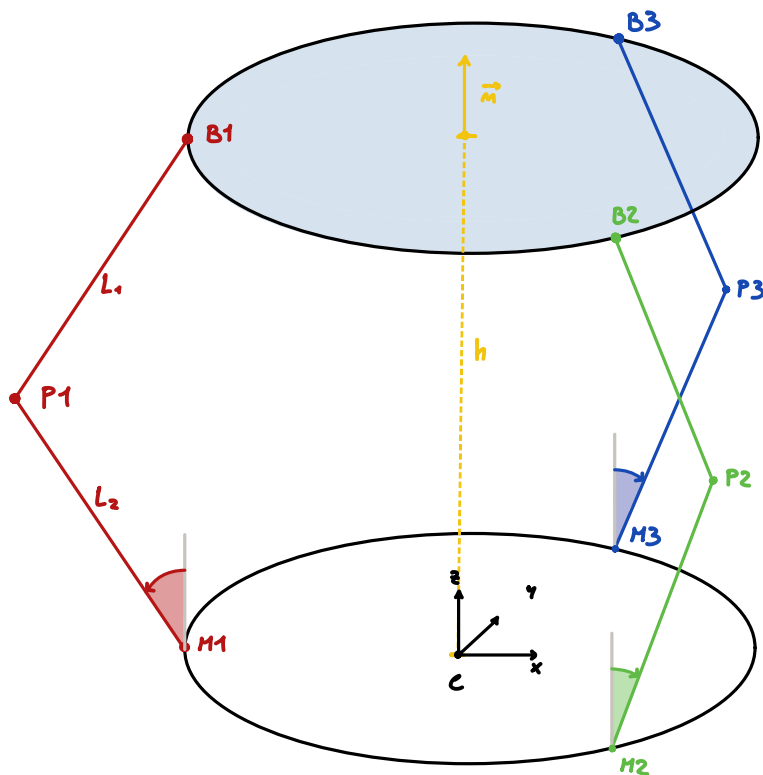




WE WILL EXPRESS THE POSE OF THE PLATFORM AS $\vec{m}$ (UNITARY NORMAL VECTOR) AND

h (z COORDINATE OF THE CENTER). WHAT ARE THE COORDINATES OF B1, B2 AND B3

AS A FUNCTION OF h AND $\vec{m}$?

$$\vec{m} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

PLANE : A POINT $P = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \text{PLANE}$ IF $P - C \perp m \Rightarrow \begin{bmatrix} x \\ y \\ z - h \end{bmatrix} \cdot \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = 0 \rightsquigarrow \boxed{\alpha x + \beta y + \gamma z - h\gamma = 0}$

$$\text{B1 : } \begin{cases} y = 0 \\ \alpha x + \gamma z = \gamma h \\ x^2 + (z - h)^2 = L^2 \end{cases} \rightsquigarrow \begin{cases} y = 0 \\ x = -\frac{\gamma(z - h)}{\alpha} \end{cases} \rightsquigarrow \frac{(z - h)^2 [1 + \gamma^2/\alpha^2]}{\alpha^2} = L^2 \rightsquigarrow z - h = \frac{L}{\sqrt{1 + (\gamma/\alpha)^2}} \rightsquigarrow z = h + \frac{L}{\sqrt{1 + (\gamma/\alpha)^2}} = h + \frac{\alpha L}{\sqrt{\alpha^2 + \gamma^2}}$$

$$x = -\frac{\gamma L}{\sqrt{\alpha^2 + \gamma^2}}$$

$$y = 0$$

$$z = h + \frac{\alpha L}{\sqrt{\alpha^2 + \gamma^2}}$$

B2 : THE PLANE PASSING THROUGH M2, C, B2 IS $y = -\sqrt{3}x$

$$\begin{cases} y = -\sqrt{3}x \\ \alpha x + \beta y + \gamma z - h\gamma = 0 \\ x^2 + y^2 + (z-h)^2 = L^2 \end{cases}$$

$$\alpha x - \beta\sqrt{3}x + \gamma z = h\gamma \rightsquigarrow z = \frac{[h\gamma - \alpha x + \beta\sqrt{3}x]}{\gamma} = h + x \frac{(\sqrt{3}\beta - \alpha)}{\gamma}$$

$$x^2 + 3x^2 + \frac{(\beta\sqrt{3}x - \alpha x)^2}{\gamma^2} = L^2$$

$$x^2 \left(4 + \frac{(\beta\sqrt{3} - \alpha)^2}{\gamma^2} \right) = L^2$$

$$x = \frac{\gamma L}{\sqrt{4\gamma^2 + (\beta\sqrt{3} - \alpha)^2}}$$

$$y = \frac{-\sqrt{3} \gamma L}{\sqrt{4\gamma^2 + (\beta\sqrt{3} - \alpha)^2}}$$

$$z = h + \frac{L (\sqrt{3}\beta - \alpha)}{\sqrt{4\gamma^2 + (\beta\sqrt{3} - \alpha)^2}}$$

B3 : THE PLANE PASSING THROUGH M3, C, B3 IS $y = \sqrt{3}x$

$$\begin{cases} y = \sqrt{3}x \\ \alpha x + \beta y + \gamma z - h\gamma = 0 \\ x^2 + y^2 + (z-h)^2 = L^2 \end{cases}$$

$$\alpha x + \beta\sqrt{3}x + \gamma z = h\gamma \rightsquigarrow z = \frac{[h\gamma - \alpha x - \beta\sqrt{3}x]}{\gamma} = h - x \frac{(\sqrt{3}\beta + \alpha)}{\gamma}$$

$$x^2 + 3x^2 + x^2 \frac{(\beta\sqrt{3} + \alpha)^2}{\gamma^2} = L^2$$

$$x^2 \left(4 + \frac{(\beta\sqrt{3} + \alpha)^2}{\gamma^2} \right) = L^2$$

$$x = \frac{\gamma L}{\sqrt{4\gamma^2 + (\beta\sqrt{3} + \alpha)^2}}$$

$$y = \frac{\sqrt{3} \gamma L}{\sqrt{4\gamma^2 + (\beta\sqrt{3} + \alpha)^2}}$$

$$z = h - \frac{L (\sqrt{3}\beta + \alpha)}{\sqrt{4\gamma^2 + (\beta\sqrt{3} + \alpha)^2}}$$

NOW WE WANT TO FIND P1, P2, P3 AS A FUNCTION OF B1, B2 AND B3.

THEY LIE ON THE SAME PLANE AS B1, M1 ; B2, M2 ; B3, M3.

$$(x - B_x)^2 + (y - B_y)^2 + (z - B_z)^2 = L_1^2$$

$$(x - M_x)^2 + (y - M_y)^2 + (z - M_z)^2 = L_2^2$$

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$$x^2 - 2B_x x + B_x^2 + y^2 - 2B_y y + B_y^2 + z^2 - 2B_z z + B_z^2 = L_1^2$$

$$x^2 - 2M_x x + M_x^2 + y^2 - 2M_y y + M_y^2 + z^2 - 2M_z z + M_z^2 = L_2^2$$

$$2x(M_x - B_x) + 2y(M_y - B_y) + 2z(M_z - B_z) - (M_x^2 - B_x^2 + M_y^2 - B_y^2 + M_z^2 - B_z^2) = L_1^2 - L_2^2$$

$$z = \frac{L_1^2 - L_2^2 - 2(M_x - B_x)x - 2(M_y - B_y)y + (M_x^2 - B_x^2 + M_y^2 - B_y^2 + M_z^2 - B_z^2)}{2(M_z - B_z)}$$

$$B = \frac{B_x - M_x}{M_z - B_z} \quad C = \frac{B_y - M_y}{M_z - B_z} \quad A = \frac{M_x^2 - B_x^2 + M_y^2 - B_y^2 + M_z^2 - B_z^2 + L_1^2 - L_2^2}{2(M_z - B_z)}$$

$$= A + Bx + Cy \leadsto \text{PLANE}$$

THE PLANES TO WHICH P, B AND M BELONG TO ARE EXPRESSED AS

$$y = Ex \quad E = 0, \sqrt{3}, -\sqrt{3}$$

$$\begin{cases} z = A + Bx + Cy \\ y = Ex \end{cases} \leadsto \begin{cases} z = A + (B + CE)x \\ y = Ex \end{cases} \leadsto \text{LINE}$$

ALONG THE LINE WE LOOK FOR THOSE Xs AT DISTANCE L₁ FROM B1:

$$L_1^2 = (x - B_x)^2 + (Ex - B_y)^2 + [(B + CE)x + (A - B_z)]^2$$

$$L_1^2 = x^2 - 2B_x x + B_x^2 + E^2 x^2 - 2EB_y x + B_y^2 + (B + CE)^2 x^2 + (A - B_z)^2 + 2(B + CE)(A - B_z)x$$

$$x^2 [1 + E^2 + (B + CE)^2] + x [-2B_x - 2EB_y + 2(B + CE)(A - B_z)] + [B_x^2 + B_y^2 + (A - B_z)^2 - L_1^2] = 0$$

$$2B_x + 2EB_y - 2(B + CE)(A - B_z) \pm \sqrt{(-2B_x - 2EB_y + 2(B + CE)(A - B_z))^2 - 4(1 + E^2 + (B + CE)^2)(B_x^2 + B_y^2 + (A - B_z)^2 - L_1^2)}$$

$$x_{1,2} = \frac{2B_x + 2EB_y - 2(B + CE)(A - B_z) \pm \sqrt{(-2B_x - 2EB_y + 2(B + CE)(A - B_z))^2 - 4(1 + E^2 + (B + CE)^2)(B_x^2 + B_y^2 + (A - B_z)^2 - L_1^2)}}{2[1 + E^2 + (B + CE)^2]}$$

